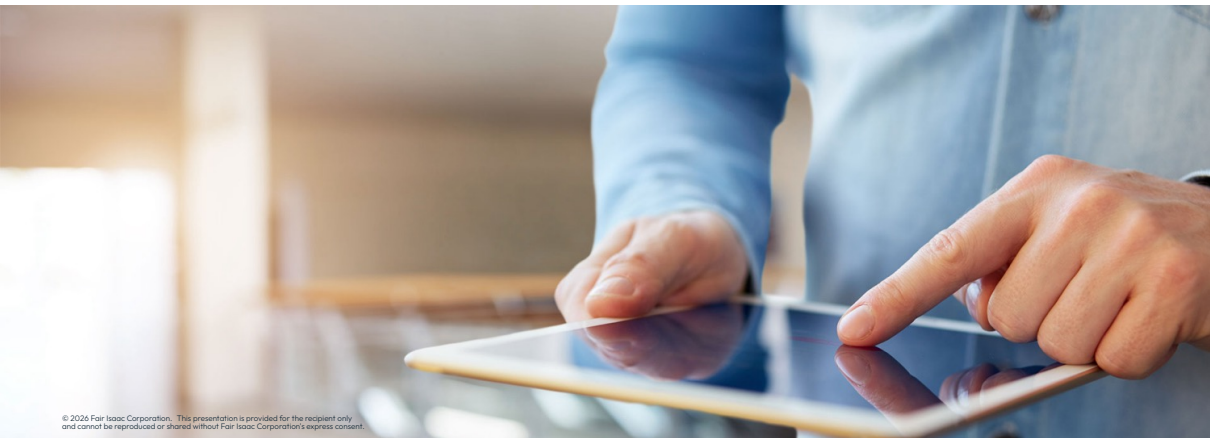


Indicator constraints





Indicator constraints

Indicator constraints

- Indicator constraints are defined by using the `problem.addIndicator()` method:

```
p.addIndicator(c1, c2, ...)
```

- An indicator constraint is a **logic constraint** that expresses the implication 'if indicator condition holds then apply the constraint':
 - Represented by a **tuple** containing a condition on a **binary variable**, called the indicator, and an expression representing a **constraint**: (indicator condition, constraint)
- Each argument `c1, c2, ...` can be a single indicator constraint, or a list, tuple, or *NumPy* array of indicator constraints (tuples)
- The constraint is only enforced when the value of the indicator variable matches a user-defined value (0 or 1)



Finding help: See also on GitHub: *indicators, unitcommitment_indicators*

Indicator constraints

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```
p.addIndicator(c1, c2, ...)
```

- Example enforcing the constraint $y \leq 15$ when binary variable $x = 1$ for an optimization problem p :

```
x = p.addVariable(vartype=xp.binary)
y = p.addVariable(lb=10, ub=20)
ind1 = (x == 1, y <= 15)
p.addIndicator(ind1)
```



Note: The `addIndicator()` method also accepts nonlinear expressions for the constraint to enforce



Thank You

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